

Shanghai Jiao Tong University
SJTU Paris Elite Institute of Technology (SPEIT)

Course Project Announcement

Optimization and Numerical Analysis

Spring 2026 | MATH6304P-260-M01

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Lecture time: Tuesday 08:00–09:40 (Weeks 2–17), Shang Yuan 204

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Major Course Project

**Development of an Automatic Design Optimization Platform for a 2D
Heat-Conduction System Using FreeFEM++**

1. Project Overview

This project concerns the development of a compact automatic design optimization platform for a two-dimensional steady-state heat-conduction problem. The project combines mathematical modeling, finite-element simulation, objective-function evaluation, optimization algorithms, and computational workflow design within a unified framework.

The required implementation is based on **FreeFEM++** for the PDE simulation component and an external optimization tool for parameter search and design improvement.

2. Project Topic

Project Title: Development of an Automatic Design Optimization Platform for a 2D Heat-Conduction System Using FreeFEM++

The project is based on the HEAT-COND benchmark, which represents a thermal design problem with heterogeneous material properties and Robin boundary conditions. The benchmark provides a suitable setting for numerical simulation, parameter optimization, and platform-level workflow integration.

3. Physical Background and Mathematical Model

The computational domain is a two-dimensional unit square containing a central spine and five pairs of fins. The benchmark geometry is shown in Figure 1 and follows the benchmark configuration reported in [1].

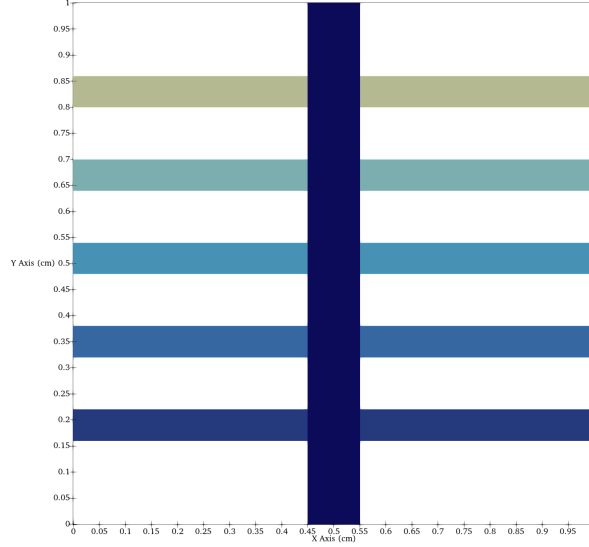


Figure 1: 2D geometry of the HEAT-COND benchmark. The unit square domain contains a central spine (width $w_s = 0.10$) and 5 pairs of fins (thickness $t_f = 0.06$) at vertical positions $y = \{0.16, 0.32, 0.48, 0.64, 0.80\}$.

The governing equation is the steady-state heat-conduction equation with mixed boundary conditions:

$$\begin{cases} -\nabla \cdot (k(\mathbf{x}) \nabla T) = 0 & \text{in } \Omega, \\ T = 1 & \text{on } \Gamma_{\text{Base}}, \\ k(\mathbf{x}) \frac{\partial T}{\partial n} + \text{Bi} T = 0 & \text{on } \Gamma_{\text{Fin}}. \end{cases} \quad (1)$$

Here:

- T denotes the dimensionless temperature,
- $k(\mathbf{x})$ denotes the piecewise-constant thermal conductivity,
- Bi denotes the dimensionless Biot number,
- Γ_{Base} denotes the base boundary,
- Γ_{Fin} denotes the fin surfaces.

The objective is to maximize the average temperature on the fin boundary:

$$J = \frac{1}{|\Gamma_{\text{Fin}}|} \int_{\Gamma_{\text{Fin}}} T \, d\Gamma. \quad (2)$$

The design variables are defined by

$$\mathbf{x} = (k_1, k_2, k_3, k_4, k_5, \text{Bi}),$$

with admissible ranges listed in Table 1.

Table 1: Design variables and admissible ranges for the HEAT-COND benchmark

Parameter	Symbol	Range
Layer conductivity	$k_i, i = 1, \dots, 5$	$[0.1, 1.0]$
Biot number	Bi	$[0.01, 1.0]$

The optimization problem is therefore written as

$$\max_{\mathbf{x}} J(\mathbf{x}). \quad (3)$$

4. Project Requirements

Each group shall develop an automatic design optimization platform including simulation, objective evaluation, optimization, and post-processing.

Task 1: PDE Model and FreeFEM++ Solver

Construct a numerical solver in FreeFEM++ for the two-dimensional steady-state heat-conduction problem in (1). The solver shall correctly handle:

- Geometry construction,
- Mesh generation,
- Region labeling,
- Piecewise material assignment,
- Dirichlet and Robin boundary conditions,
- Temperature-field output.

Task 2: Objective Evaluation Module

Implement the evaluation of the objective function in (2). For a given design vector $\mathbf{x}$, the program shall compute the corresponding average temperature along the fin boundary.

Task 3: Optimization Module

Couple the FreeFEM++ solver with an external optimization method. The optimization method may be implemented using an appropriate language or library, such as Python with `scipy.optimize`, differential evolution, particle swarm optimization, genetic algorithms, Bayesian optimization, or another suitable method.

The optimization module shall:

- Automatically transfer design variables to the simulation code,
- Execute repeated PDE solves,
- Record the optimization history,
- Report the best design obtained.

Task 4: Platform Integration

The project shall be organized as a compact automatic design optimization platform. At a minimum, the implementation shall include:

- An input/configuration module,
- A FreeFEM++ simulation module,
- An objective evaluation module,
- An optimization driver,
- A result-saving and visualization module.

Task 5: Numerical Analysis and Discussion

Each group shall provide a concise and well-structured analysis of the numerical and optimization results. The analysis may include:

- Comparison of initial and optimized temperature fields,
- Physical interpretation of the optimized parameters,
- Convergence behavior of the optimizer,
- Sensitivity to parameter choices,
- Computational cost and bottlenecks,
- Advantages and limitations of the platform design.

5. Recommended System Workflow

A recommended system workflow is:

Input Parameters → FreeFEM++ Solver → Objective Evaluation
→ Optimizer → Results and Visualization.

A reasonable implementation should allow the user to specify the design variables or optimization settings, launch the solver automatically, evaluate the objective, and store both intermediate and final results.

6. Group Policy

Students shall work in groups of **2–3 members**. Each group shall submit one final package. All members are expected to understand the overall methodology and contribute substantially to the project. **The final report shall include a brief statement describing the contribution of each member.**

7. Deliverables

7.1. Final Report

A report of approximately **6–10 pages**, written in clear academic English, which includes:

1. Introduction and project objective,
2. Mathematical model,
3. Numerical formulation and FreeFEM++ implementation,
4. Optimization strategy,
5. Numerical results,
6. Analysis and discussion,
7. Conclusion and possible improvements.

7.2. Source Code and README

Submit all source files required to reproduce the main results, including:

- FreeFEM++ files,
- Optimization scripts,

- Configuration or input files,
- A README.md describing how to run the project.

7.3. Presentation

Each group shall give a short in-class presentation introducing:

- The problem setting,
- The platform architecture,
- The numerical and optimization methods used,
- The main results and conclusions.

8. Assessment

This major project carries a total weight of **30%** of the final course grade. The grading breakdown is as follows.

Part I: Midterm Project Checkpoint (10 points out of 30)

Item	Points
Understanding of the problem and modeling clarity	3
First working version of the FreeFEM++ solver	4
Midterm progress presentation / checkpoint quality	3

Part II: Final Project (20 points out of 30)

Item	Points
Completeness and organization of the platform	6
Correctness and reliability of numerical results	5
Optimization strategy and experimental design	4
Quality of analysis and discussion	3
Writing and presentation quality	2

9. Schedule and Submission

The course is taught from **Week 2 to Week 17**. The project shall be carried out during the semester and submitted at the end of the teaching period.

Final submission deadline: the weekend of Week 17.

Intermediate checkpoints, discussion sessions, and presentation arrangements may be announced during the semester according to the progress of the course. **The professor reserves the right to adjust the deadline within a reasonable range based on the actual progress.**

10. Possible Extensions

The following extensions are encouraged, though not mandatory:

- Comparison of two or more optimization methods,
- Mesh-sensitivity analysis,
- Parameter-sensitivity analysis,
- Replacement of continuous conductivity values by a discrete material-selection framework,
- Use of surrogate models or response surfaces to accelerate optimization.

High-quality extensions may be positively considered during grading.

11. Academic Integrity

Discussion within each group is permitted. All submitted materials must represent the original work of the group. Direct copying of code, text, figures, or analysis from other groups or external sources is not permitted.

If external libraries, public repositories, online tutorials, or other resources are used, they shall be properly acknowledged in the final report. **Cases of plagiarism or academic misconduct will be handled according to university and course regulations.**

12. Final Remarks

The purpose of this project is to provide training in the combined use of numerical analysis and optimization for a representative computational engineering problem. A successful submission should demonstrate a correct numerical implementation, a coherent optimization workflow, and a clear interpretation of the results.

References

- [1] Hassan Ballout, Yvon Maday, and Christophe Prud'homme. Nonlinear compressive reduced basis approximation for multi-parameter elliptic problem. In *Multiscale, Nonlinear and Adaptive Approximation II*, pages 55–73. Springer, 2024.